

CUT Interface Specification: ThermalController-v1.0

Component: Sealed thermal chamber controller

File: `thermal_controller.p` (MATLAB pcode binary: do not attempt to open)

Module: MC3113 Mechatronic Systems Design

Function Signature

```
[outputs, new_state] = thermal_controller(inputs, state)
```

Pass `[]` as `state` on the first call. The executor (`execute_test.m`) handles this for you, you do not call `thermal_controller` directly.

Inputs (struct)

Field	Type	Unit	Description
<code>setpoint</code>	double	°C	Target chamber temperature
<code>ambient_temp</code>	double	°C	External disturbance temperature
<code>dt</code>	double	s	Timestep — always use 0.1

Outputs (struct)

Field	Type	Unit	Description
<code>chamber_temp</code>	double	°C	Simulated chamber temperature
<code>heater_cmd</code>	double	—	Heater drive command [0.0, 1.0]
<code>cooler_cmd</code>	double	—	Cooler drive command [0.0, 1.0]
<code>fault_flag</code>	int	—	0 = nominal, 1 = warning, 2 = fault

Requirements Under Test

These are the obligations the controller claims to meet.

Thresholds are NOT given here, determining them is part of your test design.

ID	Obligation
R-RESP-01	The chamber shall reach the target setpoint within a defined rise time under step stimulus
R-SS-01	Steady-state temperature error shall remain within tolerance once at setpoint

ID	Obligation
R-STAB-01	The system shall not exhibit sustained oscillation or excessive overshoot
R-CONS-01	The heater and cooler shall not operate simultaneously at any timestep
R-SAFE-01	The heater command shall be limited when chamber temperature exceeds the thermal safety threshold

What You Are NOT Told

- The PID gains
- The internal safety threshold value(s)
- The heater command cap value
- The plant time constant
- Any internal state variable meaning

Your test cases must expose the controller's behaviour, you discover these properties through systematic testing, not by reading source code.

Stimulus Types (for your Stimulus Engine)

Type	Behaviour	Key parameter
step	Setpoint jumps from ambient (20 °C) to param_setpoint at $t = 0$	param_setpoint (°C), param_duration (s)
ramp	Setpoint increases linearly from 20 °C to param_setpoint over param_duration	param_setpoint (°C), param_duration (s)
boundary	Setpoint is held at exactly param_setpoint — used to probe limit behaviour	param_setpoint (°C), param_duration (s)
sustained_fault	Setpoint drives chamber above a safety threshold and holds it	param_setpoint (°C), param_duration (s)

Notes on TC-06 and Ramp Stimulus

Important for oracle design:

rise_time_10_90 is only meaningful for **step stimulus**, where a clear 0%→100% transition exists. For **ramp stimulus**, the correct oracle metric is **ramp_tracking_lag**, the mean signed error between the actual chamber

temperature and the linearly increasing setpoint, measured once the response is past its initial transient (after the first 15 s).

If you apply `rise_time_10_90` to a ramp stimulus, the result is physically meaningless (the “10% threshold” is a moving target). Design your ramp test cases with `ramp_tracking_lag` as the metric instead.

Lab 03 Implementation Note: The Lab 03 Student Guide uses the absolute-mean variant: `measured = mean( abs( setpoint(t) - chamber_temp(t) ) ) [all t]` This is the definition to implement in `oracle.m`. The signed mean above is provided for conceptual reference only.